

SMA Sunny Central — Modbus Comms Block Mapping (Corrected)

Version: 2026-06-01 | Target: PPC_Controller [DB39], UDT *Inverter_controller*

Connection: Modbus TCP, Unit ID = 3 | Source: MODBUS-SCxxxx-TI-en-19 + SCADA Register Map XLSX

CORRECTION: Previous version incorrectly referenced GriMng.WNom, GriMng.VArNom, GriMng.PFNom — these names DO NOT EXIST in the SMA Modbus profile. Correct channels: WSpt (addr 108), VArSpt (addr 112), PFSpt (addr 114). Previous version also used wrong OpStt ENUM values (308/309) — corrected to 3526/3527.

1. Conventions

Parameter	Value
Modbus protocol	TCP/IP
Unit (Slave) ID	3
Register bit-width	32-bit — 2 consecutive 16-bit registers per value
Write function code	FC16 (0x10) Write Multiple Holding Registers
Read input registers	FC04 (0x04) — measured values / setpoint readbacks (read-only)
Read holding registers	FC03 (0x03) — writable parameters
Endianness	Big-endian (Motorola), high word first
Scan cycle	100 ms (OB30): READ input regs -> PROCESS -> WRITE holding regs / setpoints

Format	Scaling rule	Example
ENUM	Integer code (no scaling)	InvOpMod: 308 = Operation, 303 = Stop
FIX0	value / 1 (physical units)	WSpt: 2000 -> 2000 kW
FIX1	value / 10	Voltage: 4150 -> 415.0 V
FIX4	value / 10000	PFSpt: 9500 -> 0.9500

Note on WSpt/VArSpt/PFSpt addresses: These channels appear in the SMA Input Register table (FC04, read-only readback) at addresses 108/112/114. SMA Sunny Central also accepts FC16 writes to these same addresses when the corresponding control mode (WCtrlCom/VArCtlCom/PFCtlCom) is active. The FC04 readback confirms which setpoint the inverter is currently tracking.

2. WRITE to SMA Inverter — FC16 (PLC to Inverter)

Written every OB30 cycle. RemRdy must be written BEFORE InvOpMod. ErrClr is one-shot only (Section 2.3).

#	UDT Field	SMA Channel	Addr	Words	Type	Format	Unit	Written by
1	(derived)	RemRdy	2	2	S32	ENUM	-	Comms block (from OperMode)
2	OperMode	InvOpMod	0	2	S32	ENUM	-	FaultHandler
3	WMode	GriMng.WMod	6	2	S32	ENUM	-	PowerDist.
4	VArMode	GriMng.VArMod	4	2	S32	ENUM	-	ReactiveCtl
5	WSpt	WSpt	108	2	S32	FIX0	kW	PowerDist.
6	VArSpt	VArSpt	112	2	S32	FIX0	kVAr	ReactiveCtl
7	PFSpt	PFSpt	114	2	S32	FIX4	-	ReactiveCtl
8	ErrClr	ErrClr	8	2	S32	ENUM	-	FaultHandler (one-shot)

2.1 ENUM Codes

Register (Addr)	Code	Text	Meaning
InvOpMod (0)	308	Operation	Command inverter to RUN
InvOpMod (0)	303	Stop	Command inverter to STOP
RemRdy (2)	308	Ready	Grant remote operation permission
RemRdy (2)	303	Standby	Revoke remote permission
GriMng.WMod (6)	1079	WCtrlCom	Active power via Modbus — USE FOR PPC
GriMng.WMod (6)	303	Off	No active power control

GriMng.VArMod (4)	1072	VArCtlCom	Reactive power kVAr via Modbus — USE FOR PPC
GriMng.VArMod (4)	1075	PFCtlCom	Power factor via Modbus
GriMng.VArMod (4)	2270	AutoCom	Automatic — PPC selects Q or PF mode
GriMng.VArMod (4)	303	Off	No reactive control
ErrClr (8)	26	Ackn	Acknowledge fault — ONE SCAN ONLY, then write 0
ErrClr (8)	0 / 973	-	Idle

2.2 Setpoint Scaling

UDT value	Write to register	Physical meaning
WSpt = 2000	addr 108 = 2000	2000 kW active power limit
VArSpt = -500	addr 112 = -500	500 kVAr capacitive
PFSpt = 0.95 -> DInt 9500	addr 114 = 9500	cos phi = 0.950 lagging
PFSpt = 1.00 -> DInt 10000	addr 114 = 10000	cos phi = 1.000 unity

2.3 Critical Write Sequencing

START (OperMode=308): Write RemRdy=308 (reg 2) FIRST then InvOpMod=308 (reg 0) — same FC16 frame preferred.

STOP (OperMode=303): Write RemRdy=303 (reg 2) FIRST then InvOpMod=303 (reg 0).

ErrClr one-shot: Write Holding[8]=26 in ONE scan after fault cleared, then 0 in every subsequent scan. SMA requires a 0->26 rising edge. Continuous 26 has no effect.

3. READ from SMA Inverter — FC04 (Inverter to PLC)

Read every OB30 scan. On timeout or Modbus exception: set CommError=TRUE and hold last valid values. After 3 consecutive failures, block all writes to this inverter and remove from PPC dispatch.

3.1 Status and State Registers

UDT Field	SMA Channel	Addr	Words	Type	Format	Unit	Derivation / Notes
RemReady	OpStt	98	2	S32	ENUM	-	TRUE if OpStt IN {3526, 3527, 3530}
Error	ErrStt	94	2	S32	ENUM	-	TRUE if ErrStt != 307 (307=OK)
PwrOffReas	PwrOffReas	178	2	S32	ENUM	-	21626=Low Power SetPoint
DrtStt	DrtStt	176	2	S32	ENUM	-	973=no derating active
(SCADA) ErrNo	ErrNo	96	2	U32	FIX0	-	Log fault code when Error=TRUE
(SCADA) ErrLcn	ErrLcn	92	2	U32	ENUM	-	Fault location code

3.2 Measured Power Registers

UDT Field	SMA Channel	Addr	Words	Type	Format	Unit	Notes
Wactive	InvMs.TotW	28	2	S32	FIX0	kW	Measured AC active power total
Qactive	InvMs.TotVAr	30	2	S32	FIX0	kVAr	Measured AC reactive power
(disp)	InvMs.TotVA	26	2	S32	FIX0	kVA	Apparent power
(disp)	InvMs.PF	24	2	S32	FIX4	-	Power factor x10000
(disp)	GriMs.Hz	38	2	S32	FIX2	Hz	Grid frequency x100

3.3 Availability and Setpoint Readback Registers

UDT Field	SMA Channel	Addr	Words	Type	Format	Unit	Notes
WAval	WAval	172	2	S32	FIX4	pu x10000	10000 = 100% available
VArAval	VArAval	174	2	S32	FIX4	pu x10000	avail. reactive power
(readback)	WSpt	108	2	S32	FIX0	kW	Active setpoint currently in effect
(readback)	VArSpt	112	2	S32	FIX0	kVAr	Reactive setpoint currently in effect
(readback)	PFSpt	114	2	S32	FIX4	-	PF setpoint currently in effect
(commiss.)	WRtg	184	2	S32	FIX0	kW	Rated power - read once at startup

3.4 OpStt ENUM Decoding (CORRECTED)

Previous version incorrectly used OpStt=308/309. These are InvOpMod codes, NOT OpStt codes. Correct values below:

OpStt	Text	RemReady	PPC/SCADA Notes
3526	GridFeed	TRUE	Normal operation - enable closed-loop dispatch
3527	FRT	TRUE	Fault ride-through active - do not issue stop
3530	RampDown	TRUE	Controlled ramp-down - still grid-connected
3529	QonDemand	TRUE	Reactive-only mode
3528	Standby	FALSE	Not producing - restart sequence if required
1394	WaitAC	FALSE	Waiting for AC grid - check voltage/freq
1393	WaitDC	FALSE	Waiting for DC voltage
3524	ConnectAC	FALSE	AC synchronising - wait
381	Stop	FALSE	Stopped
1392	Error	FALSE	Fault - read ErrNo, acknowledge after remedy
1787	Init	FALSE	Booting - hold commands
1469	ShutDown	FALSE	Shutting down - investigate PwrOffReas
Other	-	FALSE	Unknown/transitioning

3.5 DrtStt Key Values

DrtStt Value	Text	PPC Action
973	--- (no derating)	Normal dispatch
21586	Frt	FRT active - hold setpoints, log compliance event
21601	WCtlVol	P limited by voltage - accept constraint, reduce WSpt
21651	VArPrio	P reduced for Q priority - accept, log
21591	WCtlHz	Overfrequency limitation
21590	WCtlLoHz	Underfrequency limitation

3.6 PwrOffReas Key Values

Value	Text	PPC Action
21626	Low Power SetPoint	Increase WSpt/VArSpt above inverter minimum threshold
21609	Stop: SCADA/PPC Modbus	Stop was commanded - restart only on operator request
21607	Stop: InvOpMod	InvOpMod not set to Operation - write 308
21617	Standby: RemRdy	RemRdy not Ready - write 308
21615	Standby: External Grid Error	External grid error - clear signal, restart
21612	Standby: SCADA/PPC Modbus	Standby was commanded
21613	Standby: AC Synchronization	AC grid sync issue - check V/f limits

4. Bulk Read Strategy

Step	FC	Start	Words	Registers Covered
Startup	FC03	0	104	All holding regs: modes, ramps, protection limits (read at startup/mode change)
Cycle 2a	FC04	10	106	Measurements, OpStt, ErrStt, ErrNo, WSpt/VArSpt/PFSpt readback (addrs 10-115)
Cycle 2b	FC04	116	112	WAval, VArAval, DrtStt, PwrOffReas, WRTg (addrs 116-227)

5. Per-Inverter Write Transaction — Corrected SCL Summary

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// Step 1: Fault clear (one-shot - one scan only, then 0)
IF ErrClr <> 0 THEN
    Write Holding[8] = 26      // ErrClr = Ackn
ELSE
    Write Holding[8] = 0      // ErrClr = idle
END_IF

// Step 2: Start/Stop sequence

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IF OperMode = 303 THEN          // Stop
    Write Holding[2] = 303      // RemRdy = Standby  FIRST
    Write Holding[0] = 303      // InvOpMod = Stop
ELSIF OperMode = 308 THEN      // Start
    Write Holding[2] = 308      // RemRdy = Ready    FIRST
    Write Holding[0] = 308      // InvOpMod = Operation
END_IF

// Step 3: Control modes
Write Holding[6] = WMode        // GriMng.WMod  (1079=WCtlCom, 303=Off)
Write Holding[4] = VArMode      // GriMng.VArMod (1072=VArCtlCom, 1075=PFctlCom)

// Step 4: Setpoints --- CORRECTED: NOT GriMng.WNom / GriMng.VArNom / GriMng.PFNom ---
Write [108] = WSpt              // Active power setpoint [kW, FIX0]
Write [112] = VArSpt            // Reactive setpoint   [kVAr, FIX0] if VArCtlCom
Write [114] = PFSpt            // PF setpoint           [x10000]      if PFctlCom

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6. Per-Inverter Read Transaction

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// FC04 batch read: addrs 10 to 115
Read [28] -> Wactive      (kW,  FIX0) // InvMs.TotW
Read [30] -> Qactive      (kVAr, FIX0) // InvMs.TotVAR
Read [94] -> ErrStt (ENUM) // Error := (ErrStt <> 307)
Read [96] -> ErrNo  (FIX0) // Log when Error = TRUE
Read [98] -> OpStt  (ENUM) // RemReady := OpStt IN {3526,3527,3530}
Read [108] -> WSpt readback (kW,  FIX0)
Read [112] -> VArSpt readback (kVAr, FIX0)
Read [114] -> PFSpt readback (FIX4)

// FC04 batch read: addrs 116 to 227
Read [172] -> WAval      (FIX4) // div 10000 for fraction
Read [174] -> VArAval    (FIX4)
Read [176] -> DrtStt     (ENUM) // 973=no derating
Read [178] -> PwrOffReas (ENUM) // 21626=Low Power SetPoint
Read [184] -> WRtg (kW,  FIX0) // Read once at startup

// Error handling
IF timeout OR Modbus_Exception THEN
    CommError := TRUE // Block writes; hold last valid values
END_IF
// CommError cleared on next successful read

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